

Web Development

>>HTML: Tags – Hyperlink, Media, Lists

HTML-Lists

```
graph TD; A[HTML-Lists] --> B[Unordered List]; A --> C[Description List]; A --> D[Ordered List];
```

Unordered List

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

- Coffee
- Tea
- Milk

Description List

```
<dl>
  <dt>Coffee</dt>
  <dd> black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white drink</dd>
</dl>
```

Coffee
- black hot drink
Milk
- white cold drink

Ordered List

```
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

1. Coffee
2. Tea
3. Milk

Note: List can be nested (lists inside lists)

HTML Links

- In HTML, links are defined with the `<a>` tag:

`<a href="https://www.google.com">Visit Google</a>`

- Local Links

`<a href="html_about_us.html">Company Profile</a>`

`<a href="mailto:webmaster@example.com"> email Jon Doe</a>`

It is common to use images as link

Example

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Document</title>
6 </head>
7 <body>
8   <ul>
9     <li><a href="mypage.html">Home</a></li>
10
11     <li><a href="https://www.w3schools.com/Tags/default.asp">Tags</a></li>
12
13     <li><a href="#">About us</a></li>
14
15     <li><a href="mailto:heba_zaki@adj.aast.edu">Contact us</a></li>
16   </ul>
17 </body>
18 </html>
```

← → ↻ ⓘ 127.0.0.1:55454/IntroHtml.html#

- [Home](#)
- [Tags](#)
- [About us](#)
- [Contact us](#)

HTML Links

- Create a Bookmark

HTML bookmarks are used to allow readers to jump to specific parts of a Web page.

```
<h2 id="C4">Chapter 4</h2>
```

```
<a href="#C4">Jump to Chapter 4</a>
```

Or

```
<a href="demo.html#C4">Jump external page</a>
```

HTML Images

```

```

```

```

```

```

```

```

Next-Generation Images

A collection of picture files known as "next-generation images" use cutting-edge compression techniques to decrease file size while preserving image clarity. The **JPEG 2000**, **WebP**, and **AVIF** next-generation picture standards are examples.

To convert you may follow these steps:

- **Use tools to convert your images into next-generation formats**
- **Host Your Images on Google Cloud Storage.**
- **Enable Google Cloud CDN**



Next-Generation Images (Cont.)

1- You can use tools like Adobe Photoshop or free online tools (e.g., [Squoosh](#)). For bulk conversion, you can use tools like **ImageMagick**:

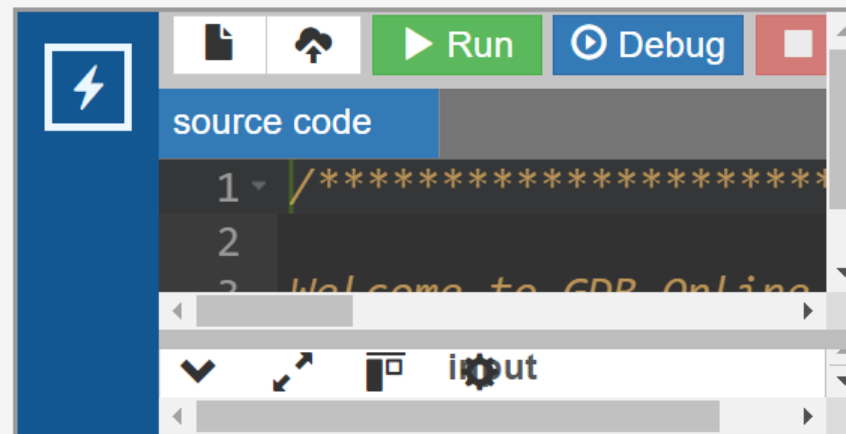
- **2- Host Your Images on Google Cloud Storage**
- Google's Cloud CDN uses **Google Cloud Storage** (GCS) as the backend, so you'll need to upload your images to a GCS bucket.
- **Create a Google Cloud Storage Bucket:**
 - Go to the Google Cloud Console.
 - Create a **Google Cloud Storage** bucket.
 - Choose the appropriate location and storage class for your images.
- **Upload Your Images:**
 - Upload the WebP/AVIF images (or both formats) into your bucket.
 - Ensure that the images have public read access:
 - Go to the **Permissions** section of the bucket.
 - Add the "allUsers" role and give them the "Storage Object Viewer" permission.

Iframes

- An **iframe** is an inline frame that allows us to embed another document within the current HTML document.

```
<iframe src="url" title="description" width="500"  
height="300"></iframe>
```

It is an example of HTML Iframe



Assignment

- Q1) What is canvas element ? Give Example
- Q2) What is SVG element? Give Example

Web Development

>> HTML: Structure Elements

This lecture is based on lectures prepared by Dr. Mohammed Misbhaudhin | Updated and Modified by Dr. Mohamed Eltaweel

Element Types in HTML

- Elements in HTML are either **Inline** or **Block**.
- **Block-level Elements**
 - Begins on a new line
 - Occupy the whole width
 - Stacks on top of one another
 - Example: <p>, <h1>.....<h6>

Element Types in HTML

- **Block-level Elements**

HTML Block Elements				
<u><address></u>	<u><article></u>	<u><aside></u>	<u><blockquote></u>	<u><canvas></u>
<u><dd></u>	<u><div></u>	<u><dl></u>	<u><dt></u>	<u><fieldset></u>
<u><figcaption></u>	<u><figure></u>	<u><footer></u>	<u><form></u>	<u><h1> - <h6></u>
<u><header></u>	<u><hr></u>	<u></u>	<u><main></u>	<u><nav></u>
<u><noscript></u>	<u></u>	<u><p></u>	<u><pre></u>	<u><section></u>
<u><table></u>	<u><tfoot></u>	<u></u>	<u><video></u>	

Element Types in HTML

- **Inline-level Elements**

- Do not begin on a new line
- Within the normal flow of the document
- Line up one after the other
- Occupy the width of the element only
- Example: `<strong>`, `<em>`

Element Types in HTML

- **Inline-level Elements**

HTML Inline Elements				
<u><a></u>	<u><abbr></u>	<u><acronym></u>	<u></u>	<u><bdo></u>
<u><big></u>	<u>
</u>	<u><button></u>	<u><cite></u>	<u><code></u>
<u><dfn></u>	<u></u>	<u><i></u>	<u></u>	<u><input></u>
<u><kbd></u>	<u><label></u>	<u><map></u>	<u><object></u>	<u><output></u>
<u><q></u>	<u><samp></u>	<u><script></u>	<u><select></u>	<u><small></u>
<u></u>	<u></u>	<u><sub></u>	<u><sup></u>	<u><textarea></u>
<u><time></u>	<u><tt></u>	<u><var></u>		

inline to block

- We can change the display of HTML element from inline to block and vice versa

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>Document</title>
6  <style>
7      ul{list-style-type: none;}
8      li {display: inline;}
9
10     </
11
12 </head>
13 <body>
14     <ul>
15
16     <li><a
17         href="https://www.google.com">
18         Download</a></li>
19     <li><a href="https://www.google.com">
20         About</a></li>
21     <li><a href="https://www.google.com">
22         Contact</a></li>
23     <li><a href="https://www.google.com">
24         Privacy</a></li>
25     <li><a href="https://www.google.com">
26         Terms</a></li>
27     <li><a href="https://www.google.com">
28         Help</a></li>
29     <li><a href="https://www.google.com">
30         Feedback</a></li>
31     <li><a href="https://www.google.com">
32         Settings</a></li>
33     <li><a href="https://www.google.com">
34         About</a></li>
35     <li><a href="https://www.google.com">
36         Privacy</a></li>
37     <li><a href="https://www.google.com">
38         Terms</a></li>
39     <li><a href="https://www.google.com">
40         Feedback</a></li>
41     <li><a href="https://www.google.com">
42         Settings</a></li>
43     <li><a href="https://www.google.com">
44         About</a></li>
45     <li><a href="https://www.google.com">
46         Privacy</a></li>
47     <li><a href="https://www.google.com">
48         Terms</a></li>
49     <li><a href="https://www.google.com">
50         Feedback</a></li>
51     <li><a href="https://www.google.com">
52         Settings</a></li>
53     <li><a href="https://www.google.com">
54         About</a></li>
55     <li><a href="https://www.google.com">
56         Privacy</a></li>
57     <li><a href="https://www.google.com">
58         Terms</a></li>
59     <li><a href="https://www.google.com">
60         Feedback</a></li>
61     <li><a href="https://www.google.com">
62         Settings</a></li>
63     <li><a href="https://www.google.com">
64         About</a></li>
65     <li><a href="https://www.google.com">
66         Privacy</a></li>
67     <li><a href="https://www.google.com">
68         Terms</a></li>
69     <li><a href="https://www.google.com">
70         Feedback</a></li>
71     <li><a href="https://www.google.com">
72         Settings</a></li>
73     <li><a href="https://www.google.com">
74         About</a></li>
75     <li><a href="https://www.google.com">
76         Privacy</a></li>
77     <li><a href="https://www.google.com">
78         Terms</a></li>
79     <li><a href="https://www.google.com">
80         Feedback</a></li>
81     <li><a href="https://www.google.com">
82         Settings</a></li>
83     <li><a href="https://www.google.com">
84         About</a></li>
85     <li><a href="https://www.google.com">
86         Privacy</a></li>
87     <li><a href="https://www.google.com">
88         Terms</a></li>
89     <li><a href="https://www.google.com">
90         Feedback</a></li>
91     <li><a href="https://www.google.com">
92         Settings</a></li>
93     <li><a href="https://www.google.com">
94         About</a></li>
95     <li><a href="https://www.google.com">
96         Privacy</a></li>
97     <li><a href="https://www.google.com">
98         Terms</a></li>
99     <li><a href="https://www.google.com">
100        Feedback</a></li>
101    </ul>
102 </body>
103 </html>
```

DIV & SPAN Tags

- Two most popular tags used in HTML without any meaning (semantics) – DIV tag and SPAN tag
- Extremely valuable when building a website
- Apply **targeted styles** to a contained set of elements

DIV Tag

- **Block-level Tag**
- Means “*Division*”
- **Purpose:** Group other elements together
- **Why used:** to apply same structure and style to them as they are related
- Use

```
<!-- Division -->
<div>
  <p>I may be found on...</p>
  <p>Additionally, I have a profile on...</p>
</div>
```

Span Tag

- **Inline-level Tag**
- **Purpose:** Small group of elements within other block-level elements
- **Why used:** To apply different style to a part of the block-level element
- Use

```
<!-- Span -->
```

```
<p>Soon we'll be <span>writing HTML</span> with the best of them.</p>
```

- *We will use this when styling in CSS*

Common Attributes in HTML

- **Class** Attribute: Assigns a common name to a group of elements. The same class name can be used for different elements

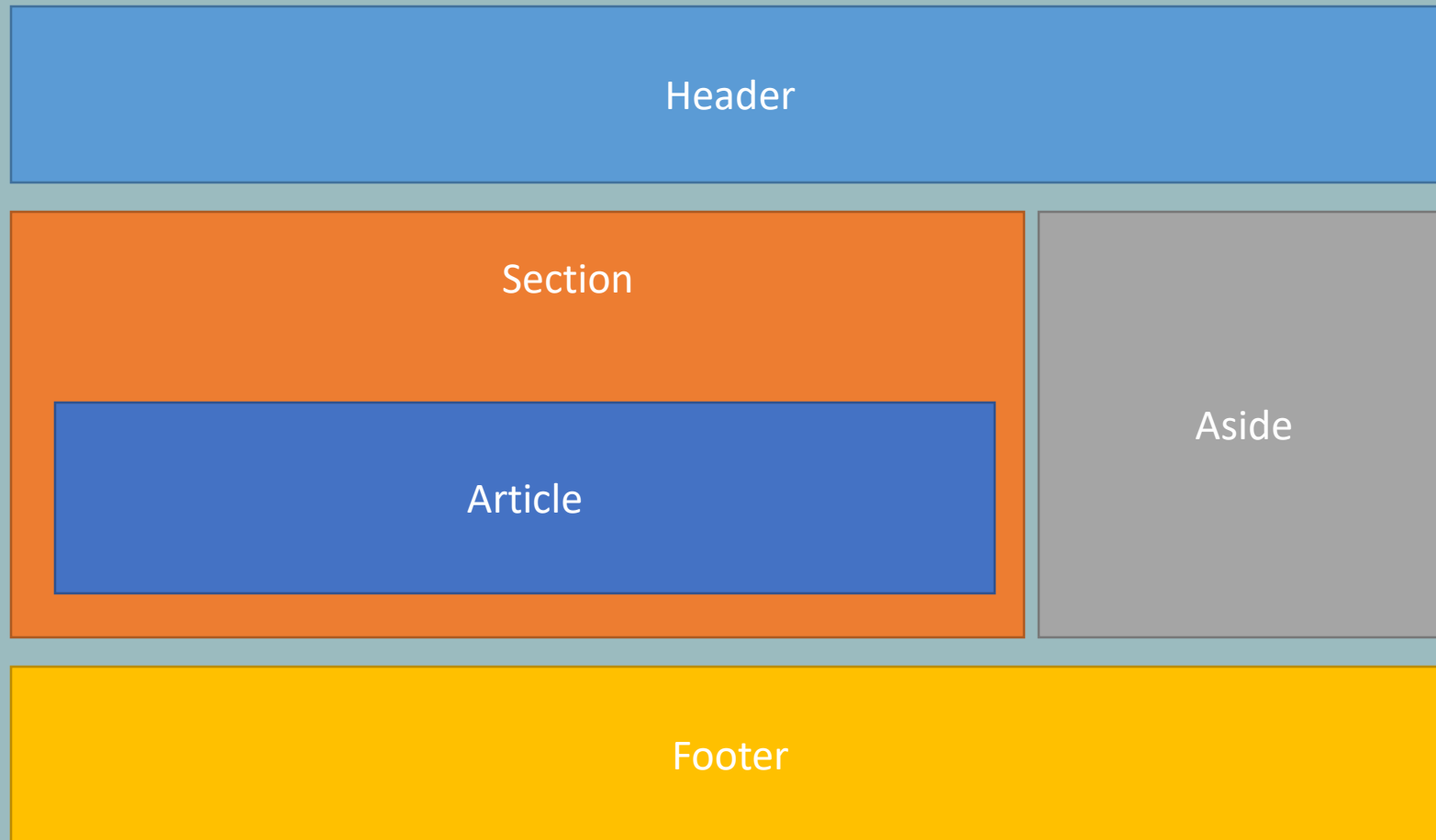
`<h1 class="heading">.....</h1>`

`<h2 class="heading">.....</h2>`

- **ID** Attribute: Assigns a unique name to an element. It cannot be repeated within the document

`<h1 id="main-heading">.....</h1>`

HTML Document Structure



Check -> HTML `<address>` Tag

HTML Structure

- Typical HTML structure was done using DIV Tags with id/class attributes

```
<body>
  <div id="header">.....</div>
  <div class="section">
    <div class="article">.....</div>
    <div class="article">.....</div>
    .....
  </div>
  <div class="section">
    <div class="article">.....</div>
    <div class="article">.....</div>
    .....
  </div>
  <div id="footer">.....</div>
</body>
```

DIV USE ISSUE

- DIV tags have no semantic value (or meaning)
- Should use class or id names to assign meaning or purpose to DIV tags
- Too many DIV tags in the document – **divitis**

HTML 5 Structure Elements

- HTML 5 handled the issue
- Introduced new structural tags
 - <header>
 - <nav>
 - <article>
 - <section>
 - <aside>
 - <footer>

Header Element

- Used to identify the top of the page or the top of a section or the top of an article
- It include introduction text such as logo, title or heading (h1 to h6 tags)
- `<header>.....</header>`

TRY NOW

- Open the page created in the previous class
- Create a header tag immediately after the body tag
- Within that tag, create a `<h1>` tag with your name

Nav Element

- Used to include the major navigation links on the page – menu bar or sitemap
- `<nav>.....</nav>`

`<nav>`

`<a href="/html/">HTML</a> |`

`<a href="/css/">CSS</a> |`

`<a href="/js/">JavaScript</a> |`

`<a href="/jquery/">jQuery</a>`

`</nav>`

[HTML](#) | [CSS](#) | [JavaScript](#) | [jQuery](#)

Section Element

- Used to group contents together on the page
- Each section may also have its own header element
- `<section>.....</section>`

TRY NOW

- Create a section after the header element
- Put the previous class code (h1 and p) inside the section
- Within the section, create a header tag and put the h2 tag inside it

Article Element

- Identify individual self-contained content on the page
- blog posts, newspaper articles, user-submitted content, and so on. `<article>.....</article>`

TRY NOW

- 
- Put all the p tags inside the article element

Aside Element

- Contain content not directly related to the main content
- Sidebar, definitions, descriptions
- `<aside>.....</aside>`

Footer Element

- Identify closing or end of page or article or section of a page
- Found at the bottom of the page/section
- `<footer>.....</footer>`

TRY NOW

- 
- Add a footer element to the page after the section element
 - Add copyright to the page © 2015 Your Name

Position and Layout

- You will see that all your elements are stacked on top of one another
- Because we did not position or layout them properly
- We will learn **Layouts** in CSS

Layout Example

```
<!DOCTYPE html>
<html>
<head>  <title>   The BIS Layout is here
        </title>
        <style>
            .Myheader{
                background-color: aquamarine;
                color: brown;
                padding: 50px;
                text-align: center;
                font-family: fantasy;
                font-size: 20px;    }
```



Layout Example

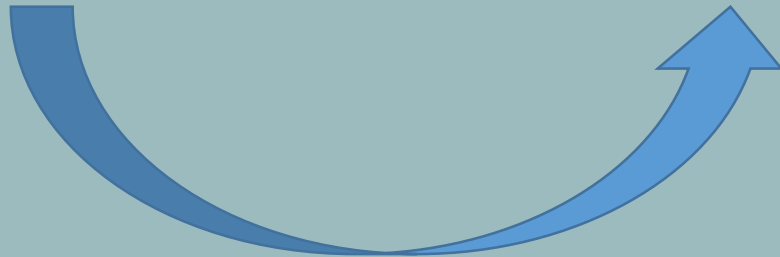
```
.sub{  
  background-color: #DDDD;  
  color: chocolate;  
  padding: 10px;  
  text-align: left;  
  font-family: sans-serif;  
  font-size: 18px;  
}
```

```
.MyFoot{ background-color: #DDDD;  
        color: chocolate;padding: 10px;  
        text-align: center;  
        font-family: fantasy;  
        font-size: 20px;}
```

```
.MyB{text-align: center;}
```

```
ul{list-style-type: none;}  
ul li{display: inline;  
padding-right: 10px;}
```

```
</style>  
</head>
```



Layout Example

```
<body>
```

```
  <div class="Myheader">
```

```
    The <span style="color: blue; font-family:
monospace;">header</span> area
```

```
    <div class="sub">
```

```
      <ul>
```

```
        <li> <button> <a href="About%20us.html">
about us</a></button> </li>
```

```
        <li><button> <a href="index.html"> Home </a>
</button> </li>
```

```
        <li> <button> <a href="http://www.google.com">
Google </a></button></li>
```

```
      </ul>
```

```
    </div>
```

```
  </div>
```

```
    <div class="MyB">
```

```
    
```

```
    <br>
```

```
      body of the website is here
```

```
    </div>
```

```
    <div class="MyFoot">The copy rights
are reserved to AYA &copy; and the logo
is &reg;</div>
```

```
  </body>
```

```
</html>
```



Summary

- DIV and Span Tags
- ID and Class Attributes
- HTML 5 Structure Tags
 - Header, Nav, Section, Article, Aside and Footer

Reference for all active and obsolete tages

<https://developer.mozilla.org/en-US/docs/Web/HTML/Element/tt>